THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 12: **Beyond Earth** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which is the correct order of size (largest to smallest) among Jupiter, Earth, Mars, Mercury? 1

(a) Jupiter, Earth, Mars, Mercury **(b)** Mercury, Mars, Earth, Jupiter
(c) Earth, Jupiter, Mars, Mercury **(d)** Jupiter, Mars, Earth, Mercury

2. A planet that has a day shorter than that of the Earth would: 1

(a) rotate faster **(b)** rotate slower **(c)** not rotate **(d)** orbit faster

3. Which of these is a limitation of observing the sky from the Earth's surface? 1

(a) atmosphere blurs the view **(b)** planets are not visible
(c) Moon blocks it **(d)** stars are too bright

4. A Moon rock sample brought back is studied to learn about: 1

(a) cooking **(b)** the Moon's history
(c) weather **(d)** farming

5. Which of these statements about the Earth's axis is true? 1

(a) it points to the Sun
(b) it points to the Moon
(c) Pole star lies almost along its extension
(d) it changes daily

6. Why are space telescopes placed outside the atmosphere? 1

(a) to be near the Sun **(b)** to cool them
(c) clear view without air disturbance **(d)** to save fuel

7. Why is it important to track space debris? 1

(a) it provides light **(b)** it brings rain
(c) it is useful fuel **(d)** it can damage satellites

8. Which of the following is the best reason why a star seen at night may not be present in the sky during the day? 1

(a) the star moves **(b)** the star dies
(c) the Sun's bright light hides it **(d)** the star is gone

9. A rover on Mars sends data that takes about 12 minutes to reach Earth. This is because: 1

(a) radio waves take time to travel the distance
(b) Earth is blocked
(c) the rover is slow
(d) Mars is a star

10. Which of the following is a renewable power source for a satellite? 1

(a) coal **(b)** petrol **(c)** wood **(d)** solar panels

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is Venus's surface hotter than Mercury's though it is farther from the Sun? 2

12. Why do we get day and night on the Earth but not seasons on the Moon in the same way? 2

13. What causes the twinkling of stars but not of planets? 2

14. Explain why the Pole star will not be exactly overhead at the equator. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how an eclipse differs from the phases of the Moon. 3

16. Explain why a satellite does not fall to Earth. 3

17. Why does the Moon look larger near the horizon? 3

OR

Why is it easier to launch spacecraft from the Moon than from the Earth? 3

18. How do astronomers find the age of stars (idea)? 3

OR

A spacecraft goes to Mars at 25000 km/h. Roughly how long to travel 75 million km? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how the knowledge of the solar system developed from naked-eye observations to space probes. 5

OR

Describe how the Sun's energy is used on the Earth and from satellites through solar panels. 5

20. A satellite's orbit time is 2 hours and its altitude is 20,000 km. Describe the reasoning for choosing orbits for communication, weather and navigation. 5

OR

Argue for or against continuing space exploration despite its cost, giving five points. 5

END OF QUESTION PAPER — Total: 40 Marks